

```

Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
Control-C
PS C:\Users\sidha> ping 172.19.0.5

Pinging 172.19.0.5 with 32 bytes of data:
Request timed out.

Ping statistics for 172.19.0.5:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
Control-C
PS C:\Users\sidha> ping 172.19.0.6

Pinging 172.19.0.6 with 32 bytes of data:
Request timed out.

Ping statistics for 172.19.0.6:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
Control-C
PS C:\Users\sidha>

64 bytes from 172.19.0.4: icmp_seq=33 ttl=64 time=0.175 ms
64 bytes from 172.19.0.4: icmp_seq=34 ttl=64 time=0.073 ms
64 bytes from 172.19.0.4: icmp_seq=35 ttl=64 time=0.061 ms
64 bytes from 172.19.0.4: icmp_seq=36 ttl=64 time=0.075 ms
64 bytes from 172.19.0.4: icmp_seq=37 ttl=64 time=0.090 ms
64 bytes from 172.19.0.4: icmp_seq=38 ttl=64 time=0.077 ms
64 bytes from 172.19.0.4: icmp_seq=39 ttl=64 time=0.079 ms
64 bytes from 172.19.0.4: icmp_seq=40 ttl=64 time=0.066 ms
64 bytes from 172.19.0.4: icmp_seq=41 ttl=64 time=0.110 ms
64 bytes from 172.19.0.4: icmp_seq=42 ttl=64 time=0.087 ms
64 bytes from 172.19.0.4: icmp_seq=43 ttl=64 time=0.080 ms
64 bytes from 172.19.0.4: icmp_seq=44 ttl=64 time=0.071 ms
64 bytes from 172.19.0.4: icmp_seq=45 ttl=64 time=0.087 ms
64 bytes from 172.19.0.4: icmp_seq=46 ttl=64 time=0.061 ms
64 bytes from 172.19.0.4: icmp_seq=47 ttl=64 time=0.065 ms
64 bytes from 172.19.0.4: icmp_seq=48 ttl=64 time=0.087 ms
64 bytes from 172.19.0.4: icmp_seq=49 ttl=64 time=0.063 ms
64 bytes from 172.19.0.4: icmp_seq=50 ttl=64 time=0.084 ms

64 bytes from 172.19.0.4: icmp_seq=26 ttl=64 time=0.078 ms
64 bytes from 172.19.0.4: icmp_seq=27 ttl=64 time=0.071 ms
64 bytes from 172.19.0.4: icmp_seq=28 ttl=64 time=0.068 ms
64 bytes from 172.19.0.4: icmp_seq=29 ttl=64 time=0.058 ms
64 bytes from 172.19.0.4: icmp_seq=30 ttl=64 time=0.061 ms
64 bytes from 172.19.0.4: icmp_seq=31 ttl=64 time=0.075 ms
64 bytes from 172.19.0.4: icmp_seq=32 ttl=64 time=0.072 ms
64 bytes from 172.19.0.4: icmp_seq=33 ttl=64 time=0.068 ms
64 bytes from 172.19.0.4: icmp_seq=34 ttl=64 time=0.072 ms
64 bytes from 172.19.0.4: icmp_seq=35 ttl=64 time=0.067 ms
64 bytes from 172.19.0.4: icmp_seq=36 ttl=64 time=0.084 ms
64 bytes from 172.19.0.4: icmp_seq=37 ttl=64 time=0.072 ms
64 bytes from 172.19.0.4: icmp_seq=38 ttl=64 time=0.082 ms
64 bytes from 172.19.0.4: icmp_seq=39 ttl=64 time=0.067 ms
64 bytes from 172.19.0.4: icmp_seq=40 ttl=64 time=0.052 ms
64 bytes from 172.19.0.4: icmp_seq=41 ttl=64 time=0.076 ms
64 bytes from 172.19.0.4: icmp_seq=42 ttl=64 time=0.081 ms
64 bytes from 172.19.0.4: icmp_seq=43 ttl=64 time=0.078 ms

34.418916 172.19.0.4 -> 172.19.0.5 ICMP Echo (ping) reply
35.089425 172.19.0.6 -> 172.19.0.4 ICMP Echo (ping) request
35.089448 172.19.0.4 -> 172.19.0.6 ICMP Echo (ping) reply
35.484305 172.19.0.5 -> 172.19.0.4 ICMP Echo (ping) request
35.484327 172.19.0.4 -> 172.19.0.5 ICMP Echo (ping) reply
34.146297 172.19.0.6 -> 172.19.0.4 ICMP Echo (ping) request
34.146321 172.19.0.4 -> 172.19.0.6 ICMP Echo (ping) reply
34.687631 172.19.0.5 -> 172.19.0.4 ICMP Echo (ping) request
34.687662 172.19.0.4 -> 172.19.0.5 ICMP Echo (ping) reply
35.147687 172.19.0.6 -> 172.19.0.4 ICMP Echo (ping) request
35.147719 172.19.0.4 -> 172.19.0.6 ICMP Echo (ping) reply
35.673216 172.19.0.5 -> 172.19.0.4 ICMP Echo (ping) request
35.673249 172.19.0.4 -> 172.19.0.5 ICMP Echo (ping) reply
36.148866 172.19.0.6 -> 172.19.0.4 ICMP Echo (ping) request
36.148890 172.19.0.4 -> 172.19.0.6 ICMP Echo (ping) reply
36.738646 172.19.0.5 -> 172.19.0.4 ICMP Echo (ping) request
36.738671 172.19.0.4 -> 172.19.0.5 ICMP Echo (ping) reply
37.150119 172.19.0.6 -> 172.19.0.4 ICMP Echo (ping) request

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Figure 6: Experiment 1 – Basic connectivity check

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Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
Control-C
PS C:\Users\sidha> ping 172.19.0.5

Pinging 172.19.0.5 with 32 bytes of data:
Request timed out.

Ping statistics for 172.19.0.5:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
Control-C
PS C:\Users\sidha> ping 172.19.0.6

Pinging 172.19.0.6 with 32 bytes of data:
Request timed out.

Ping statistics for 172.19.0.6:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
Control-C
PS C:\Users\sidha>

64 bytes from 172.19.0.4: icmp_seq=54 ttl=64 time=0.098 ms
64 bytes from 172.19.0.4: icmp_seq=55 ttl=64 time=0.141 ms
64 bytes from 172.19.0.4: icmp_seq=56 ttl=64 time=0.094 ms
64 bytes from 172.19.0.4: icmp_seq=57 ttl=64 time=0.070 ms
64 bytes from 172.19.0.4: icmp_seq=58 ttl=64 time=0.092 ms
64 bytes from 172.19.0.4: icmp_seq=59 ttl=64 time=0.077 ms
64 bytes from 172.19.0.4: icmp_seq=60 ttl=64 time=0.081 ms
64 bytes from 172.19.0.4: icmp_seq=61 ttl=64 time=0.072 ms
64 bytes from 172.19.0.4: icmp_seq=62 ttl=64 time=0.075 ms
64 bytes from 172.19.0.4: icmp_seq=63 ttl=64 time=0.065 ms
64 bytes from 172.19.0.4: icmp_seq=64 ttl=64 time=0.066 ms
64 bytes from 172.19.0.4: icmp_seq=65 ttl=64 time=0.073 ms
64 bytes from 172.19.0.4: icmp_seq=66 ttl=64 time=0.094 ms
64 bytes from 172.19.0.4: icmp_seq=67 ttl=64 time=0.080 ms
64 bytes from 172.19.0.4: icmp_seq=68 ttl=64 time=0.132 ms
^C--- 172.19.0.4 ping statistics ---
69 packets transmitted, 69 packets received, 0% packet loss
round-trip min/avg/max/stddev = 0.051/0.082/0.175/0.023 ms
#

msf6 > use auxiliary/scanner/portscan/syn
msf6 auxiliary(scanner/portscan/syn) > set INTERFACE eth0
INTERFACE => eth0
msf6 auxiliary(scanner/portscan/syn) > set PORTS 80
PORTS => 80
msf6 auxiliary(scanner/portscan/syn) > set RHOSTS 172.19.0.4/30
RHOSTS => 172.19.0.4/30
msf6 auxiliary(scanner/portscan/syn) > set THREADS 50
THREADS => 50
msf6 auxiliary(scanner/portscan/syn) > run
SIOCSIFFLAGS: Operation not permitted
/usr/share/metasploit-framework/lib/msf/core/exploit/capture.rb:123: warning
: undefining the allocator of T_DATA class PCAPRUB::Pcap
[+] TCP OPEN 172.19.0.4:80
[+] TCP OPEN 172.19.0.6:80
[*] Scanned 4 of 4 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/portscan/syn) > |

sh-3.2# ifconfig
eth0      Link encap:Ethernet  HWaddr 0a:c0:ea:4a:11:10
          inet addr:172.19.0.4  Bcast:172.19.255.255  Mask:255.255.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:403 errors:0 dropped:0 overruns:0 frame:0
          TX packets:307 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:38087 (37.1 KB)  TX bytes:31764 (31.0 KB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:289 errors:0 dropped:0 overruns:0 frame:0
          TX packets:289 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:135449 (132.2 KB)  TX bytes:135449 (132.2 KB)

sh-3.2#

```

Figure 7: Experiment 2 - Exploitation of Metasploitable from Kali container

each other within the Docker network and are received at Metasploitable, as indicated by the continuous ICMP replies between the internal IP addresses. This demonstrates that the containers are fully interconnected and

can communicate with one another, but remain isolated from the host system, ensuring the testbed's security and containment as illustrated in Figure 7.

The top right panel in the figure shows the exploitation of

Metasploitable images with the IP address 172.19.0.4 from the Kali Linux image with the IP address 172.19.0.5 using one of the auxiliary scanner modules of Metasploit. SYN port scan was run from Kali